ON MODULI SPACES OF UNIFORMLY NEGATIVELY CURVED METRICS

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Abstract. Let $\mathrm{Neg}(X)$ denote the space of complete Riemannian metrics with uniformly negative curvature on a surface X, equipped with the intrinsic uniform C^2 topology. Let $\mathcal{M}(X) = \mathrm{Neg}(X)/\mathrm{Diff}(X)$ be the corresponding moduli space, with the quotient topology. We construct elementary locally constant functionals on $\mathcal{M}(X)$, with values in finite symmetric products of $\{0,1\}$, based on geodesic string counts. As an upshot we show that $\mathcal{M}(\mathbb{R}\times S^1)$ is disconnected. This is perhaps surprising: the two metrics we separate are joined by an explicit path of metrics with constant curvature -1. The point is that this path is only continuous in the weak Whitney topology. More generally, if X is a finite-type surface of hyperbolic type with n punctures, then the pure moduli space has at least 2^n connected components, while $\mathcal{M}(X)$ has at least $n+1$ connected components.

1. Introduction

Spaces and moduli spaces of negatively curved metrics have been studied before, especially for closed manifolds, see for instance [7], [6]. For the analogous questions for spaces of metrics with positive scalar curvature and with non-negative curvature, see [8] and [9], respectively. The noncompact surface case with the intrinsic uniform topology seems to be a different and less explored setting. Our main thrust is to prove disconnection results for the moduli spaces of metrics on surfaces. Our approach is exceptionally simple: we use Gauss–Bonnet and Arzelà–Ascoli theorems to establish compactness of certain closed geodesic spaces, and using this construct locally constant geodesic counting functionals. At the same time the idea shadows a deeper theory involving the Fuller index, which we will indicate.

Let $\mathrm{Neg}(X)$ denote the space of Riemannian, complete metrics g with uniformly negative curvature on a surface X. By uniformly negative we mean that there is a constant $a > 0$ such that the Gaussian curvature satisfies $K_g \leq -a$ on all of X. The space $\mathrm{Neg}(X)$ will be given the intrinsic uniform C^2 topology, see Definition 2.1. This is a metrizable topology, finer than the weak Whitney C^2_{loc} topology but coarser than the strong Whitney C^2 topology. The corresponding moduli space is

$$\mathcal{M}(X) = \mathrm{Neg}(X)/\mathrm{Diff}(X),$$

with the quotient topology, where diffeomorphisms act by pullback. If X has distinguished punctures, we also write

$$\mathcal{M}_{\mathrm{p}}(X) = \mathrm{Neg}(X)/\mathrm{Diff}_{\mathrm{p}}(X),$$

where $\mathrm{Diff}_{\mathrm{p}}(X)$ denotes the subgroup preserving each puncture, or equivalently each end, individually.

This moduli space should be distinguished from the usual Riemann moduli space $\mathcal{M}_{g,n}$. By uniformization, the latter may be viewed as the moduli space of complete finite-area hyperbolic metrics, so all punctures are cusp ends. In the present paper the ambient moduli space is larger: the same underlying smooth punctured surface is allowed to carry uniformly negatively curved metrics with either cusp or funnel type behavior at a puncture. The intrinsic uniform topology remembers this large-scale end geometry. Thus the all-cusp hyperbolic locus lies in one sector of our space, while the geodesic string counting invariant below separates the other cusp/funnel sectors. This should be compared with the usual Teichmüller topology on the all-cusp hyperbolic locus.

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Theorem 1.1. *The moduli space*

$$\mathcal{M}(\mathbb{R} \times S^1) = \mathrm{Neg}(\mathbb{R} \times S^1)/\operatorname{Diff}(\mathbb{R} \times S^1)$$

admits a natural nonconstant, locally constant function and so is disconnected.

Corollary 1.2. *The moduli space*

$$\mathcal{M}^{\mathrm{strong}}(\mathbb{R} \times S^1) = \mathrm{Neg}^{\mathrm{strong}}(\mathbb{R} \times S^1)/\operatorname{Diff}(\mathbb{R} \times S^1)$$

is disconnected, where the strong superscript means that we take the strong Whitney C^2 topology.

Proof. Since the strong Whitney topology is finer, this is immediate. □

The role of the topology. The disconnection in Theorem 1.1 is genuinely a feature of the intrinsic uniform topology. The two metrics g_0, g_1 exhibited in Section 3, with $N(g_0, \beta) = 0$ and $N(g_1, \beta) = 1$, are the warped products

$$g_i = dx^2 + f_i(x)^2\, d\theta^2,$$

with

$$f_0(x) = e^x, \qquad f_1(x) = \cosh x.$$

Interpolating the warping functions linearly,

$$f_s(x) = (1-s)e^x + s\cosh x, \qquad s \in [0,1],$$

one has $f_s'' = f_s$, and therefore

$$K_{g_s} = -\frac{f_s''}{f_s} = -1.$$

Thus each $g_s = dx^2 + f_s(x)^2 d\theta^2$ lies in $\mathrm{Neg}(\mathbb{R} \times S^1)$. The path $s \mapsto g_s$ is continuous in the weak Whitney (C^∞_{loc}) topology, so the two metrics lie in one weak Whitney path component; in particular their images in $\mathcal{M}(\mathbb{R} \times S^1)$ lie in one path component for the quotient of the weak Whitney topology.

At the marked level the particular metrics in this example are also not globally bi-Lipschitz, and so the order-zero part of the topology already separates them. For this reason the applications below are stated on the moduli spaces. A diffeomorphism of a non-compact surface need not preserve a global bi-Lipschitz class, while the invariant constructed here is natural under diffeomorphisms.

We now describe a basic metric deformation invariant. A *geodesic string* in X, g will be short for: an equivalence class of a constant speed smooth g-geodesic mapping $S^1 \to X$, under the S^1 reparametrization action. Let $\pi_1^*(X)$ be the set of non-trivial free homotopy classes of loops in X (topologized with the discrete topology). Define

$$N : \mathrm{Neg}(X) \times \pi_1^*(X) \to \{0, 1\},$$

by declaring $N(g, \beta)$ to be the number of class β g-geodesic strings. This number is at most one by the negative curvature assumption.

Theorem 1.3. *The function N is locally constant. It is natural under diffeomorphisms: if $\phi : X \to X$ is a diffeomorphism, then*

$$N(\phi^* g, \beta) = N(g, \phi_* \beta).$$

Finally,

$$N(g, \beta) = N(g, \beta^{-1}).$$

Consequently, for every subgroup $G \subset \operatorname{Diff}(X)$, N descends to a locally constant function

$$N_G : (\mathrm{Neg}(X) \times \pi_1^*(X))/G \to \{0, 1\}.$$

Here G acts diagonally by

$$(g, \beta) \cdot \phi = (\phi^* g, (\phi^{-1})_* \beta).$$

In particular, if $\mathcal{B} \subset \pi_1^*(X)$ is a finite subset which is preserved by G up to permutation and inversion, then the unordered collection of values defines a locally constant function

$$\mathcal{N}_{\mathcal{B},G} : \mathrm{Neg}(X)/G \to \mathrm{Sym}^{|\mathcal{B}|}(\{0,1\}), \qquad \mathcal{N}_{\mathcal{B},G}([g]) = \{N(g,\beta)\}_{\beta\in\mathcal{B}}.$$

If $|\mathcal{B}| = 1$, we regard $\mathcal{N}_{\mathcal{B},G}$ as a $\{0,1\}$-valued function.

A deeper theory in the background. Let $\mathrm{Met}(X)$ denote the space of complete Riemannian metrics with uniformly negatively curved ends. In forthcoming work, we construct an extension of N to a locally constant, $\mathrm{Diff}(X)$ natural functional:

$$F : \mathrm{Met}(X) \times \pi_1^*(X) \to \mathbb{Q},$$

built using Fuller index theory, but still based on the same idea of counting geodesic strings. This requires a more involved analogue of the compactness Theorem 2.3, and is beyond the methods of the present paper. Granting it, we would get that $\mathrm{Met}(\mathbb{R} \times S^1)/\,\mathrm{Diff}(\mathbb{R} \times S^1)$ is disconnected.

Going back to the theorem above, taking $\mathcal{B}$ to be the peripheral classes around the ends, the local constancy of the functions $\mathcal{N}_{\mathcal{B},G}$ gives the following finite-type moduli statement.

Theorem 1.4. *Let X be an orientable finite-type surface without boundary, with $n \geq 1$ punctures, and assume that X is of hyperbolic type, i.e. $\chi(X) < 0$. Then the pure moduli space*

$$\mathcal{M}_{\mathrm{p}}(X) = \mathrm{Neg}(X)/\,\mathrm{Diff}_{\mathrm{p}}(X)$$

has at least 2^n connected components, and the full moduli space

$$\mathcal{M}(X) = \mathrm{Neg}(X)/\,\mathrm{Diff}(X)$$

has at least $n + 1$ connected components.

2. The space of geodesic strings and N

In what follows all metrics g are elements of $\mathrm{Neg}(X)$, with the latter as in the Introduction. Free homotopy classes of loops correspond to conjugacy classes of deck transformations of the universal cover $\widetilde{X} \to X$. For $\beta \in \pi_1^*(X)$ let $\eta_\beta \neq \mathrm{id}$ be a deck transformation in the conjugacy class determined by β. The translation length

$$|\eta_\beta|_g = \inf_{\widetilde{x}\in\widetilde{X}} \widetilde{d}(\widetilde{x}, \eta_\beta\widetilde{x})$$

used below is conjugation invariant, so depends only on β.

Let LX denote the space of all smooth maps $S^1 \to X$,

$$S^1 = \mathbb{R}/\mathbb{Z},$$

with the usual C^0 topology. The space $S(g,\beta) \subset LX$ will denote the subspace of all constant speed parametrized, g-geodesics in class β. Set

$$\mathcal{O}(g,\beta) = S(g,\beta)/S^1,$$

the quotient by the S^1 action, where S^1 is acting by reparametrization: $t\cdot o(\tau) = o(\tau+t)$. The elements of $\mathcal{O}(g,\beta)$ are called geodesic strings.

Since the metrics in $\mathrm{Neg}(X)$ have strictly negative curvature, there is at most one class β geodesic string. Indeed, two geometrically distinct such geodesics would lift to axes of the same deck transformation in the Hadamard universal cover, and the flat strip theorem would give a flat strip, impossible under negative curvature. We therefore define

$$N(g,\beta) = \#\mathcal{O}(g,\beta) \in \{0,1\}.$$

Theorem 1.3 says that this elementary count is a locally constant deformation invariant.

We give $\mathrm{Neg}(X)$ the intrinsic uniform C^2 topology, which is very natural in our setup, and particularly in the compactness argument of Theorem 2.3.

Definition 2.1. *The intrinsic uniform C^2 topology is defined as follows. A neighbourhood basis of $g \in \mathrm{Neg}(X)$ consists, for $\lambda > 1$ and $\epsilon > 0$, of the sets*

$$N_{g,\lambda,\epsilon} = \Big\{ h \in \mathrm{Neg}(X) \;\Big|\; \sup_{x\in X} \sup_{0\neq v \in T_x X} \Big| \log \frac{h_x(v,v)}{g_x(v,v)} \Big| < \log \lambda,\ \sup_{x \in X} \big(|\nabla^g (h-g)|_g + |(\nabla^g)^2 (h-g)|_g \big) < \epsilon \Big\},$$

where ∇^g, $|\cdot|_g$ are the Levi-Civita connection and tensor norms of g. That is, h is uniformly λ-bi-Lipschitz to g at order 0, and uniformly ϵ-close at orders 1 and 2, measured in g itself. This is a metrizable topology [5], finer than the weak Whitney C^2_{loc} topology (coarser than the strong Whitney topology). The order-0 condition is multiplicative on purpose: it controls the length functional uniformly over all of X, including along curves whose images leave every compact set, and is the only order used in Lemma 2.4. Order 1 controls the Christoffel symbols, hence which curves are nearly geodesic, see Lemma 2.8. Order 2 controls the curvature, hence the geodesic flow and its linearisation, which is what the persistence of closed geodesics in Theorem 1.3 uses. No derivative of order ≥ 3 is used anywhere.

For a fixed metric the compactness we are after is elementary, and it is instructive to isolate why: a class β geodesic is automatically a minimiser, so it can neither collapse nor, under the uniform negative curvature assumption, spread over a non-compact region. Theorem 2.3 is the assertion that this confinement survives passage to a compact family of metrics.

Lemma 2.2. *Fix $g \in \mathrm{Neg}(X)$. Then every class β g-geodesic string has length equal to the translation length*

$$|\eta_\beta|_g := \inf_{\widetilde{x} \in \widetilde{X}} \widetilde{d}(\widetilde{x}, \eta_\beta \widetilde{x}),$$

and is a minimiser of length in β; in particular all such geodesic strings have the same length, and

$$\inf_{o \in \beta} \ell_g(o) = |\eta_\beta|_g,$$

where

$$\ell_g(o) = \int_{S^1} \sqrt{g(\dot{o}, \dot{o})}.$$

Moreover the images of all elements of $S(g,\beta)$ are contained in a single compact subset of X.

Proof. Let $\pi : \widetilde{X} \to X$ be the universal cover, a Hadamard manifold by Cartan–Hadamard. That a closed geodesic in a non-positively curved manifold minimises length in its free homotopy class β, with length equal to the translation length $|\eta_\beta|_g$ is well known; it follows from the standard correspondence between such geodesics and axes of η_β and the convexity of the displacement function on a Hadamard manifold, see for instance [2, II.6]. In particular all class β geodesics have the common length $|\eta_\beta|_g$ and $\inf_{o\in\beta} \ell_g(o) = |\eta_\beta|_g$.

For the confinement, there is nothing to prove if $S(g,\beta)$ is empty. If there is only one geometric class β geodesic, its image is compact. Otherwise choose two geometrically distinct class β geodesics γ_0, γ_1, with lifts $\widetilde{\gamma}_0, \widetilde{\gamma}_1$. By the preceding paragraph, these axes have equal translation length, hence are parallel, and by the flat strip theorem [4, Proposition 5.1] they bound a flat, totally geodesic strip in $\widetilde{X}$. This is impossible because $K_g \leq -a < 0$ on all of X, for some $a > 0$. Hence there is at most one class β geodesic string, and the images of all elements of $S(g,\beta)$ are contained in a single compact subset of X. $\square$

Given a subset $\mathscr{K} \subset \mathrm{Neg}(X)$ set

$$\mathcal{O}(\mathscr{K}, \beta) := \{ (o, g) \in (LX/S^1) \times \mathscr{K} \mid o \in \mathcal{O}(g,\beta) \}.$$

Theorem 2.3. *Let $\mathscr{K} \subset \mathrm{Neg}(X)$ be compact. Then the space $\mathcal{O}(\mathscr{K},\beta)$ is compact.*

The proof uses a few lemmas. The first is the continuity of the infimal length; it is exactly here that the uniform, rather than weak Whitney, topology is used.

Lemma 2.4. *The infimal length*

$$\mathcal{L}(g) = \inf_{o\in\beta} \ell_g(o) \tag{2.5}$$

defines a finite continuous function on $\mathrm{Neg}(X)$*.*

Proof. Fixing any smooth $o_\beta \in \beta$ gives $\mathcal{L}(g) \le \ell_g(o_\beta) < \infty$. For the continuity, fix g. By Definition 2.1, for every $\delta > 0$ there is a neighbourhood V of g such that, for all $h \in V$,

$$(1+\delta)^{-1} g(v,v) \le h(v,v) \le (1+\delta) g(v,v) \qquad \text{for all } x \in X,\ v \in T_xX. \tag{2.6}$$

Taking square roots and integrating along any loop o gives

$$(1+\delta)^{-1}\ell_g(o) \le \ell_h(o) \le (1+\delta)\ell_g(o),$$

with a bound independent of o. Since this estimate holds for every $o \in \beta$, it passes to the infimum:

$$(1+\delta)^{-1}\mathcal{L}(g) \le \mathcal{L}(h) \le (1+\delta)\mathcal{L}(g), \qquad h \in V.$$

Hence $\mathcal{L}$ is continuous at g. Note that no closed geodesic need exist for this argument: only near-minimisers are used, so it applies verbatim where the infimum (2.5) is not attained. ☐

The next lemma records the only fact about the limit metric needed below. It gives a positive lower length bound, but does not assert that this lower bound is attained.

Lemma 2.7. *Let* $g_j \to g'$ *in* $\mathrm{Neg}(X)$*, and let* $b_j \in S(g_j, \beta)$*. Then*

$$\mathcal{L}(g') = \inf_{o\in\beta} \ell_{g'}(o) > 0.$$

Proof. By Lemma 2.2, b_j minimises g_j-length in the class β. Hence, for every loop $o \in \beta$,

$$\mathcal{L}(g_j) = \ell_{g_j}(b_j) \le \ell_{g_j}(o).$$

For j large, g_j and g' are uniformly bi-Lipschitz close, so

$$\ell_{g_j}(o) \le (1+\delta)\ell_{g'}(o)$$

for all $o \in \beta$. Taking the infimum over o gives

$$\mathcal{L}(g_j) \le (1+\delta)\mathcal{L}(g').$$

Since b_j is a genuine closed geodesic, $\mathcal{L}(g_j) > 0$. If $\mathcal{L}(g')$ were zero, the preceding inequality would force $\mathcal{L}(g_j) = 0$ for all large j, a contradiction. Hence $\mathcal{L}(g') > 0$. ☐

The second lemma quantifies the elementary fact that a geodesic of a nearby metric is nearly geodesic; this is where order 1 of Definition 2.1 is used.

Lemma 2.8. *Let* $g, h \in \mathrm{Neg}(X)$ *with* $h \in N_{g,\lambda,\epsilon}$ *as in Definition 2.1, and let b be a closed h-geodesic. Then the g-geodesic curvature of b satisfies*

$$|\kappa_g(b)| \le C(\lambda)\epsilon,$$

where $C(\lambda)$ *depends only on* λ*.*

Proof. The difference of Levi-Civita connections $\Lambda := \nabla^g - \nabla^h$ is a tensor, given by the standard formula

$$2h(\Lambda(u,v),w) = -(\nabla^g_u h)(v,w) - (\nabla^g_v h)(u,w) + (\nabla^g_w h)(u,v),$$

and $\nabla^g h = \nabla^g(h-g)$. Using the bi-Lipschitz comparison between $|\cdot|_g$ and $|\cdot|_h$ this gives

$$|\Lambda|_g \leq C(\lambda) \sup_X |\nabla^g(h-g)|_g \leq C(\lambda)\epsilon.$$

If b is an h-geodesic then $\nabla^g_{\dot b}\dot b = \Lambda(\dot b, \dot b)$, so

$$|\kappa_g(b)| \leq \frac{|\nabla^g_{\dot b}\dot b|_g}{|\dot b|_g^2} \leq |\Lambda|_g \leq C(\lambda)\epsilon.$$

□

The third lemma is the main geometric element of the argument, an approximate form of the flat strip theorem. It is proved by passing to the cyclic cover of X associated to the subgroup $\langle \eta_\beta \rangle \subset \pi_1(X)$, and using Gauss–Bonnet.

Lemma 2.9. *Let (X,g) be a complete surface with $K_g \leq 0$, and let $b_0, b_1 : S^1 \to X$ be disjoint loops in the same non-trivial free homotopy class β. Suppose that, for $i = 0,1$, the loop b_i has g-length $< D$, has g-geodesic curvature $< \delta_i$, and has a lift (as a map of $\mathbb{R}$) to the universal cover that is an embedded line. Let $\widehat{\Omega}$ be the compact annulus in the cyclic cover cobounded by the corresponding lifts $\widehat{b}_0, \widehat{b}_1$ and A its area. Then there is a point $x \in \widehat{\Omega}$ with*

$$K_g(x) > -\epsilon(A, D, \delta_0, \delta_1),$$

where

$$\lim_{\delta_0, \delta_1 \to 0} \epsilon(A, D, \delta_0, \delta_1) = 0.$$

Here K_g denotes the curvature of the lifted metric on the cyclic cover.

Proof. Let $p : \widehat{X} \to X$ be the cover associated to $\langle \eta_\beta \rangle$. With the pulled-back metric, p is a local isometry and $\widehat{X}$ is an infinite cylinder. The loops b_0, b_1 lift to closed homotopic loops $\widehat{b}_0, \widehat{b}_1$ in $\widehat{X}$.

By hypothesis the lift $\widetilde{b}_i$ of b_i to the universal cover is embedded. Hence $\widehat{b}_0$ and $\widehat{b}_1$ are embedded circles in $\widehat{X}$. Explicitly, fix η_β-equivariant lift $\widetilde{b}_i$ of b_i, then in the quotient by the $\langle \eta_\beta \rangle$ action this gives $\widehat{b}_i$, and it is embedded.

Since the original loops are disjoint, the two embedded loops $\widehat{b}_0, \widehat{b}_1$ are disjoint and cobound a compact sub-annulus $\widehat{\Omega}$.

Gauss–Bonnet on the cobounding annulus. By Gauss–Bonnet,

$$\int_{\widehat{\Omega}} K_g \, dA + \int_{\partial \widehat{\Omega}} \kappa_g \, ds = 2\pi \chi(\widehat{\Omega}) = 0.$$

The boundary curves have lengths $< D$ and geodesic curvatures $< \delta_0$, $< \delta_1$, as p is a local isometry. Therefore

$$\left| \int_{\partial \widehat{\Omega}} \kappa_g \, ds \right| < D(\delta_0 + \delta_1),$$

and consequently

$$\left| \int_{\widehat{\Omega}} K_g \, dA \right| < D(\delta_0 + \delta_1). \tag{2.10}$$

Since $K_g \leq 0$ on $\widehat{\Omega}$ we have

$$0 \leq -\int_{\widehat{\Omega}} K_g \, dA < D(\delta_0 + \delta_1).$$

If

$$K_g(x) \le -\frac{D(\delta_0+\delta_1)}{A} \qquad \text{for every } x \in \widehat{\Omega},$$

then

$$-\int_{\widehat{\Omega}} K_g\, dA \ge D(\delta_0+\delta_1),$$

contradicting the previous strict inequality. Hence there is a point $x \in \widehat{\Omega}$ such that

$$K_g(x) > -\frac{D(\delta_0+\delta_1)}{A}.$$

Taking

$$\epsilon(A, D, \delta_0, \delta_1) = \frac{D(\delta_0+\delta_1)}{A},$$

we are done. □

Proof of Theorem 2.3. Since the topology on $\mathrm{Neg}(X)$ is metrizable, it suffices to prove sequential compactness of

$$S(\mathscr{K}, \beta) := \{(o, g) \in LX \times \mathscr{K} \mid o \in S(g, \beta)\},$$

with $\mathcal{O}(\mathscr{K}, \beta)$ being the S^1 quotient of the latter. Let

$$(b_j, g_j) \in S(\mathscr{K}, \beta)$$

be a sequence. Passing to a subsequence, we may assume that $g_j \to g' \in \mathscr{K}$. We show that the corresponding loops have a C^0-convergent subsequence.

First, the lengths are controlled with respect to the fixed metric g'. Choose a smooth loop $o_\beta \in \beta$. Since $g_j \to g'$, the metrics g_j and g' are uniformly bi-Lipschitz for all large j, and

$$\ell_{g_j}(b_j) = \mathcal{L}(g_j) \le \ell_{g_j}(o_\beta)$$

is bounded. Enlarging the constant, we therefore have

$$\ell_{g'}(b_j) < D \tag{2.11}$$

for all large j. Since the b_j are parametrised with constant g_j-speed and g_j is uniformly bi-Lipschitz to g', the family is equicontinuous with respect to the distance of g'.

If the images of the b_j are contained in a fixed compact subset of X, Arzelà–Ascoli gives a C^0-convergent subsequence. The geodesic equation, together with the uniform C^2 convergence $g_j \to g'$ on this compact set, implies that the limit is a smooth g'-geodesic in the same free homotopy class. Thus it remains only to rule out escape to infinity.

Suppose, for contradiction, that after passing to a subsequence the images of the b_j leave every compact subset of X. By Lemma 2.8 and the convergence $g_j \to g'$,

$$\delta_j := |\kappa_{g'}(b_j)|_{C^0} \to 0. \tag{2.12}$$

By Lemma 2.7, $\mathcal{L}(g') > 0$. Thus every loop in the class β has g'-length at least $\mathcal{L}(g')$. Since $g' \in \mathrm{Neg}(X)$, there is $a > 0$ such that

$$K_{g'} \le -a$$

on all of X.

The g'-diameters of the curves b_j are bounded by D. Since their images leave every compact set, after passing to a subsequence we may choose indices $j < k$ so that

$$d_{g'}(b_j, b_k) \ge 1.$$

We also choose j, k so large that δ_j and δ_k are small.

Let $p : \widehat{X} \to X$ be the cyclic cover associated to $\langle \eta_\beta \rangle$, and let $\widehat{\Omega}$ be the compact annulus cobounded by the corresponding lifts $\widehat{b}_j$ and $\widehat{b}_k$, in the class $\widehat{\beta}$. Let ρ be the distance from $\widehat{b}_j$ in the lifted metric, restricted to $\widehat{\Omega}$; it is 1-Lipschitz, vanishes on $\widehat{b}_j$, and on $\widehat{b}_k$ is at least $d_{g'}(b_j, b_k) \geq 1$. Hence for almost every $t \in (0,1)$ the level $\rho^{-1}(t)$ has an embedded S^1 component separating the two boundary circles, which projects to a class β loop and so has length at least $\mathcal{L}(g')$. By the coarea formula and $|\nabla\rho| \leq 1$,

$$A = \text{area}(\widehat{\Omega}) \geq \int_0^1 \ell_{g'}\big(\rho^{-1}(t)\big)\, dt \geq \mathcal{L}(g').$$

The η_β-equivariant lifts of b_j and b_k are embedded lines, since they are geodesics for the Hadamard metrics $\widetilde{g}_j$ and $\widetilde{g}_k$. Thus the hypotheses of Lemma 2.9, applied to the metric g' and the loops b_j, b_k, are satisfied. Since

$$\epsilon(A, D, \delta_j, \delta_k) \leq \frac{D(\delta_j + \delta_k)}{\mathcal{L}(g')},$$

we may take j, k so large that this number is $< a$. The lemma gives a point $x \in \widehat{\Omega}$ with

$$K_{g'}(x) > -a,$$

contradicting $K_{g'} \leq -a$ on all of X. This rules out escape to infinity, and hence proves compactness. $\square$

Descent to the moduli space uses that $\text{Diff}(X)$ acts on $\text{Neg}(X)$ by homeomorphisms of the intrinsic topology, in fact isometrically for the structure defining it.

Lemma 2.13. *For every $\phi \in \text{Diff}(X)$ the pullback $\phi^* : \text{Neg}(X) \to \text{Neg}(X)$ is a homeomorphism. More precisely, for every $g \in \text{Neg}(X)$ and all $\lambda > 1$, $\epsilon > 0$,*

$$\phi^*\big(N_{g,\lambda,\epsilon}\big) = N_{\phi^* g,\, \lambda,\, \epsilon}. \tag{2.14}$$

Proof. The map $\phi : (X, \phi^* g) \to (X, g)$ is a Riemannian isometry by construction, hence affine and norm preserving: for every tensor T and every $k \geq 0$,

$$(\nabla^{\phi^* g})^k(\phi^* T) = \phi^*\big((\nabla^g)^k T\big), \qquad |\phi^* T|_{\phi^* g} = |T|_g \circ \phi.$$

In particular $K_{\phi^* g} = K_g \circ \phi$, so $\phi^* g$ is complete with $K_{\phi^* g} \leq \sup_X K_g < 0$, and ϕ^* maps $\text{Neg}(X)$ into itself. Applying these identities to $T = h - g$ at orders $k = 1, 2$, and writing the order-zero ratio at x in terms of $y = \phi(x)$ and $w = d\phi_x v$, each of the pointwise quantities entering Definition 2.1 for the pair $(\phi^* h, \phi^* g)$ at x equals the corresponding quantity for (h, g) at $\phi(x)$; as v ranges over $T_x X$, the vector w ranges over $T_{\phi(x)} X$. Since ϕ is a bijection of X, taking the supremum over X preserves each condition, giving (2.14). As $(\phi^{-1})^*$ has the same form, ϕ^* is a homeomorphism. $\square$

Proof of Theorem 1.3. We now prove local constancy. Fix $\beta \in \pi_1^*(X)$ and $g \in \text{Neg}(X)$. Suppose first that $N(g, \beta) = 0$. If $N(\cdot, \beta)$ were not locally zero near g, then there would be metrics $g_j \to g$ and class β geodesic strings $o_j \in \mathcal{O}(g_j, \beta)$. The set

$$\{g\} \cup \{g_j\}_{j \geq 1}$$

is compact in $\text{Neg}(X)$. By Theorem 2.3, after passing to a subsequence the strings o_j converge to a class β g-geodesic string, contradicting $N(g, \beta) = 0$. Thus $N(\cdot, \beta)$ is locally zero near g.

Now suppose that $N(g, \beta) = 1$, and let o be the unique class β geodesic string. Since $K_g < 0$, it has no non-trivial periodic normal Jacobi field, and is therefore non-degenerate, as is well known. By the implicit function theorem for non-degenerate closed geodesics, the geodesic string o persists uniquely for all metrics sufficiently close to g. Thus $N(\cdot, \beta) = 1$ on a neighbourhood of g.

This proves that

$$N : \text{Neg}(X) \times \pi_1^*(X) \to \{0, 1\}$$

is locally constant.

Naturality is immediate. A diffeomorphism $\phi : X \to X$ identifies the $\phi^* g$-geodesic strings in the class β with the g-geodesic strings in the class $\phi_*\beta$. Hence

$$N(\phi^* g, \beta) = N(g, \phi_*\beta).$$

Reversing the orientation of a closed geodesic identifies the class β strings with the class β^{-1} strings, so

$$N(g, \beta) = N(g, \beta^{-1}).$$

For the diagonal action in the statement,

$$(g, \beta) \cdot \phi = (\phi^* g, (\phi^{-1})_*\beta),$$

the naturality formula gives

$$N(\phi^* g, (\phi^{-1})_*\beta) = N(g, \beta).$$

Therefore N descends to

$$N_G : (\mathrm{Neg}(X) \times \pi_1^*(X))/G \to \{0, 1\}.$$

The quotient map is open, since it is the quotient by a group action by homeomorphisms (Lemma 2.13). Thus the descent of a locally constant invariant is again locally constant.

The assertion about the functions $\mathcal{N}_{\mathcal{B},G}$ is the same statement applied to a finite collection of classes, using $N(g, \beta) = N(g, \beta^{-1})$ when a diffeomorphism reverses a class. □

3. Some calculations

Let $X = \mathbb{R} \times S^1$, and let β be the class of the loop $\gamma(t) = (0, t)$. For a positive function f on $\mathbb{R}$, consider the warped product metric

$$g_f = dx^2 + f(x)^2 d\theta^2.$$

Here $d\theta^2$ is the standard metric on $S^1 = \mathbb{R}/\mathbb{Z}$; for background on warped products, see [1]. For such a metric the Gauss curvature is

$$K_{g_f} = -\frac{f''}{f}. \tag{3.1}$$

Also, the circle $x = x_0$ is a geodesic if and only if

$$f'(x_0) = 0. \tag{3.2}$$

Indeed, the only relevant Christoffel symbol is

$$\Gamma^x_{\theta\theta} = -ff',$$

so a circle $x = x_0$ is geodesic exactly when $f'(x_0) = 0$. Every loop in the generator class is freely homotopic to one of these circles, and the length of the circle $x = x_0$ is proportional to $f(x_0)$.

Now set

$$f_0(x) = e^x, \qquad f_1(x) = \cosh x,$$

and

$$g_0 = dx^2 + e^{2x} d\theta^2, \qquad g_1 = dx^2 + \cosh^2(x) d\theta^2.$$

Since $f_0'' = f_0$ and $f_1'' = f_1$, equation (3.1) gives

$$K_{g_0} = K_{g_1} = -1.$$

Thus both metrics belong to $\mathrm{Neg}(X)$.

For g_0, one has $f_0'(x) = e^x > 0$ for all x. Hence there is no closed geodesic in the generator class, and so

$$N(g_0, \beta) = 0.$$

For g_1, one has $f_1'(x) = \sinh x$, so the unique circle satisfying (3.2) is $x = 0$. Thus g_1 has a single class β geodesic string, represented by $t \mapsto (0, t)$. Therefore

$$N(g_1, \beta) = 1.$$

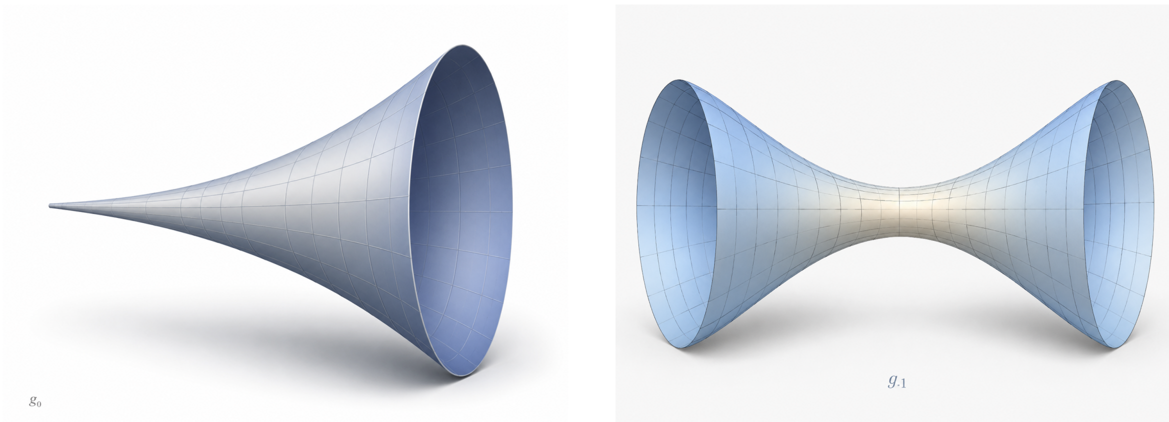

FIGURE 1. Schematic renderings of the two warped products. On the left, $g_0 = dx^2 + e^{2x} d\theta^2$, with no class β geodesic. On the right, $g_1 = dx^2 + \cosh^2(x) d\theta^2$, with a unique class β geodesic at the waist $x = 0$.

The weak Whitney path described in the introduction is especially simple in these coordinates. It is given by

$$f_s(x) = (1 - s)e^x + s \cosh x, \qquad g_s = dx^2 + f_s(x)^2 d\theta^2.$$

Since $f_s'' = f_s$, every g_s has curvature -1.

4. The cylinder quotient

Proof of Theorem 1.1. Let β be the generator of $\pi_1(\mathbb{R} \times S^1)$, and put

$$G = \mathrm{Diff}(\mathbb{R} \times S^1), \qquad \mathcal{B} = \{\beta\}.$$

Every diffeomorphism sends β to β or β^{-1}, so Theorem 1.3 gives a locally constant $\{0, 1\}$-valued function

$$\mathcal{N}_{\mathcal{B},G} : \mathcal{M}(\mathbb{R} \times S^1) \to \{0, 1\}.$$

Since $\mathcal{N}_{\mathcal{B},G}$ is locally constant, its fibers are both open and closed.

The metrics g_0, g_1 of Section 3 satisfy

$$\mathcal{N}_{\mathcal{B},G}([g_0]) = 0, \qquad \mathcal{N}_{\mathcal{B},G}([g_1]) = 1.$$

Therefore $\mathcal{M}(\mathbb{R} \times S^1)$ contains two non-empty disjoint clopen sets, and is disconnected. $\square$

5. Finite-type surfaces

Proof of Theorem 1.4. Let $p_1, \dots, p_n$ be the punctures of X, and let β_i be the primitive peripheral class going around p_i. We first construct, for each vector

$$\varepsilon = (\varepsilon_1, \dots, \varepsilon_n) \in \{0, 1\}^n,$$

a metric $g_\varepsilon \in \mathrm{Neg}(X)$ such that

$$N(g_\varepsilon, \beta_i) = \varepsilon_i. \tag{5.1}$$

Choose a pair-of-pants decomposition of a compact core of X, whose external cuffs correspond to the punctures. For the external cuff corresponding to p_i, prescribe length 0 if $\varepsilon_i = 0$, and length 1 if

$\varepsilon_i = 1$; here length 0 means a cusp. Choose one positive length for each internal cuff, assigning the same length to the two copies which are to be glued. Hyperbolic pairs of pants with these prescribed boundary lengths, allowing length 0 for cusps, exist; see for instance [3, Chapter 3]. Gluing the internal cuffs with matching lengths gives a hyperbolic compact core. At each external cuff of positive length, attach the standard hyperbolic funnel of the same boundary length. The resulting complete metric has curvature -1 everywhere, hence belongs to $\mathrm{Neg}(X)$.

If $\varepsilon_i = 0$, then the class β_i is parabolic for the hyperbolic metric g_ε, so it has no closed geodesic representative and

$$N(g_\varepsilon, \beta_i) = 0.$$

If $\varepsilon_i = 1$, then β_i is represented by the core geodesic of the attached funnel. This representative is unique. Thus

$$N(g_\varepsilon, \beta_i) = 1.$$

This proves (5.1).

For the pure diffeomorphism group, set

$$G_{\mathrm{p}} = \mathrm{Diff}_{\mathrm{p}}(X), \qquad \mathcal{B}_i = \{\beta_i\}.$$

Each $\mathcal{B}_i$ is preserved by G_{p} up to inversion, so Theorem 1.3 gives locally constant functions

$$\mathcal{N}_{\mathcal{B}_i, G_{\mathrm{p}}} : \mathcal{M}_{\mathrm{p}}(X) \to \{0,1\}.$$

Equivalently, the map

$$\overline{\Phi}([g]) = \big(\mathcal{N}_{\mathcal{B}_1, G_{\mathrm{p}}}([g]), \dots, \mathcal{N}_{\mathcal{B}_n, G_{\mathrm{p}}}([g])\big)$$

is a continuous map

$$\overline{\Phi} : \mathcal{M}_{\mathrm{p}}(X) \to \{0,1\}^n.$$

The metrics g_ε realise all binary vectors in $\{0,1\}^n$. Thus the corresponding fibers of $\overline{\Phi}$ give at least 2^n non-empty disjoint clopen subsets. Hence the pure quotient has at least 2^n connected components.

For the full diffeomorphism group, a diffeomorphism may permute the punctures. Thus the ordered vector need not descend, but its unordered version does. Set

$$G = \mathrm{Diff}(X), \qquad \mathcal{B} = \{\beta_1, \dots, \beta_n\}.$$

The set $\mathcal{B}$ is preserved by G up to permutation and inversion, and Theorem 1.3 gives

$$\mathcal{N}_{\mathcal{B}, G} : \mathcal{M}(X) \to \mathrm{Sym}^n(\{0,1\}).$$

In particular the function

$$\overline{N}([g]) = \#\{i \mid N(g, \beta_i) = 1\}$$

is locally constant and well-defined on the full quotient:

$$\overline{N} : \mathcal{M}(X) \to \{0,1,\dots,n\}.$$

The construction above realises each value $N = 0, 1, \dots, n$, and so the full quotient has at least $n+1$ connected components. □

References

[1] R. L. Bishop and B. O'Neill, *Manifolds of negative curvature*, Trans. Am. Math. Soc., 145 (1969), pp. 1–49.

[2] M. R. Bridson and A. Haefliger, *Metric Spaces of Non-Positive Curvature*, vol. 319 of Grundlehren der mathematischen Wissenschaften, Springer, 1999.

[3] P. Buser, *Geometry and Spectra of Compact Riemann Surfaces*, vol. 106 of Progress in Mathematics, Birkhäuser, 1992.

[4] P. Eberlein and B. O'Neill, *Visibility manifolds*, Pac. J. Math., 46 (1973), pp. 45–109.

[5] J. Eichhorn, *Spaces of Riemannian metrics on open manifolds*, Results in Mathematics, 27 (1995), pp. 256–283.

[6] F. T. Farrell and P. Ontaneda, *The Teichmüller space of pinched negatively curved metrics on a hyperbolic manifold is not contractible*, Annals of Mathematics (2), 170 (2009), pp. 45–65.

[7] ———, *On the topology of the space of negatively curved metrics*, Journal of Differential Geometry, 86 (2010), pp. 273–301.

[8] B. Hanke, T. Schick, and W. Steimle, *The space of metrics of positive scalar curvature*, Publ. Math., Inst. Hautes Étud. Sci., 120 (2014), pp. 335–367.

[9] W. Tuschmann and M. Wiemeler, *On the topology of moduli spaces of non-negatively curved riemannian metrics*, Mathematische Annalen, 384 (2022), pp. 1629–1651.

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